Disrupting the Industrial Metaverse Using Open Private 5G and Edge

5G is seemingly the next big thing of the current decade, so it is about time the world adapts to its offerings and moves ahead along with it. Let us try to understand how it will affect the ongoing industrial revolution or Industry 4.0.

Today, manufacturing is the first thought that comes to our minds when we look at an industrial setup that predominantly consists of mining and power grid. In order to digitise the industry entirely you need to ensure that all the Internet of Things (IoT) devices, robots, and drones across the entire country are connected. You need to run real-time applications to enable the strict need for industrial applications, hence, creating a need for reliable connectivity that supports high bandwidth and low latency at the same time.

Let us look at a few reasons for requiring these applications:

Enhanced mobile broadband:

- 20/10 Gbps DL/UL
- 4ms user plane latency
- Mobility up to 500km/hr

Ultra-reliable low latency communication:

- 1ms user plane latency
- Highly secure/resilient
- 0ms mobile interruption time/ always available

Massive machine-type communication:

- 1 million devices per km square
- 10+ years battery life
- 20dB coverage enhancement

Private 5G and edge cloud may disrupt emerging technologies

A Wi-Fi network is most suitable for the indoors, but it is not scalable when you have to cover it on a 10 square kilometre area. To do that you will have to put a Wi-Fi modem at every 10 to 20 metres, which is impractical and costly at the same time. In this case, a cellular network is the most appropriate option.

We already have working 4G networks. These networks are